Course Name:	Web Development Laboratory	**Semester:**	**III**
Date of Performance:	3/8/2026	**DIV/ Batch No:**	A2
Student Name:	Kehan Desai	**Roll No:**	16010125050

Experiment No: 1

Title

Design and Development of an AI-based Smart Irrigation Advisory Web Portal Front-End using HTML5

Aim of the Experiment

To design and develop a multi-page, use-case-based web portal front-end using HTML5 for an AI-based Smart Irrigation Advisory System for Sugarcane Crop (KJS-AGR-01), demonstrating proper HTML5 document structure, semantic elements, formatting, lists, hyperlinks, images, image maps, tables, forms, and multimedia elements.

Objectives of the Experiment

- To develop structured and semantic HTML5 web pages for an AI-enabled smart irrigation advisory system.
- To create interactive web pages for farmer registration, irrigation recommendations, weather information, crop monitoring, and advisory services.
- To apply a real-world AI and IoT-based agriculture use case (Smart Irrigation Advisory System) as the context for front-end web development.
- To build a complete HTML5 front-end portal that can later be enhanced using CSS3, JavaScript, PHP, and React.

COs to be achieved

CO1: Developing webpages using HTML and CSS.

Books / Journals / Websites references

1. *(Students should write — e.g., MDN Web Docs HTML5 reference, W3C.)*

Theory

HTML5 is the latest version of HyperText Markup Language used to create structured and interactive web pages. It introduces semantic elements such as <header>, <nav>, <main>,

<section>, <article>, and <footer>, making web pages more organized, accessible, and search-engine friendly.

HTML5 also provides support for multimedia elements like <audio> and <video>, interactive forms, tables, hyperlinks, image maps, and embedded content through <iframe>. These features help developers create user-friendly web applications for various domains such as healthcare, education, business, and agriculture.

In this experiment, HTML5 is used to design the front-end of an AI-based Smart Irrigation Advisory System. The portal enables farmers to register their farms, view irrigation schedules, monitor weather conditions, access crop information, receive AI-generated irrigation recommendations, and visualize farm locations. The application serves farmers, agricultural officers, researchers, and sugar factory management by providing organized and easy-to-understand agricultural information.

Problem Statement

"AI-based Smart Irrigation Advisory Web Portal"

An agricultural organization wants to develop a basic HTML5-based web portal front-end for an **AI-based Smart Irrigation Advisory System**. The portal should display farm information, soil moisture status, weather forecasts, irrigation recommendations, crop information, and advisory services for farmers, agricultural officers, researchers, and sugar factory management.

Each task below contributes one page or feature of this portal, representing different modules of the Smart Irrigation Advisory System.

Task 1: Create the Homepage Structure

Scenario

Design the homepage of the **Irrigation Advisory System**.

Requirements

- Use proper HTML5 document structure tags:
  - <header>, <nav>, <section>, <article>, <aside>, <footer>
- Header must contain:
  - System Name in <h1>
  - Tagline: *"Smart Irrigation using AI & Sensor Technology"*
- Navigation menu with links:
  - Home
  - Farm Dashboard
  - Irrigation Advisory
  - Sensor Data

    - Contact
- Footer with copyright information.

Concepts Tested

- HTML5 document structure
- Semantic tags
- Headings
- Hyperlinks

Deliverable: index.html

Task 2: Display Farm Information using Lists

Scenario

Display important farm information and irrigation resources.

Requirements

Create:

- Ordered List for
    - Irrigation Workflow
- Unordered List for
    - Sensors Used
        - Soil Moisture Sensor
        - Temperature Sensor
        - Humidity Sensor
        - Rainfall Sensor
- Description List for
    - Crop Stages
        - Germination
        - Tillering
        - Grand Growth
        - Maturity

Use formatting tags:

- <b>
- <i>
- <mark>

Concepts Tested

- Lists

- Formatting tags
- Text formatting

Deliverable: farm-info.html

Task 3: Irrigation Advisory Page using Formatting Tags

Scenario

Display today's irrigation recommendation for a sugarcane field.

Requirements

Display:

- Plot ID
- Farmer Name
- Soil Moisture
- Recommended Irrigation Duration
- Water Stress Level

Use:

-
-
- <mark>
- <small>
-
- <hr>

Concepts Tested

- Text formatting
- Block-level elements
- Inline formatting

Deliverable: advisory.html

Task 4: Farm Gallery using Images

Scenario

Create a gallery showing different farm conditions.

Requirements

Display four images:

- Healthy Sugarcane Crop
- Drip Irrigation System
- Soil Moisture Sensor
- Weather Monitoring Station

Each image should:

- Open detailed information when clicked
- Include:
  - alt
  - title
  - width
  - height

Use:

- <figure>
- <figcaption>

Concepts Tested

- Images
- Hyperlinks
- Global attributes

Deliverable: gallery.html

Task 5: Farm Plot Navigation using Image Map

Scenario

Provide navigation to different farm plots.

Requirements

Display a farm map.

Create clickable areas for:

- Plot A
- Plot B
- Plot C

Each area should open the corresponding plot details.

Use:

- <map>

- <area>

Concepts Tested

- Image Maps
- Hyperlinks

Deliverable: plot-map.html

Task 6: Soil Moisture & Irrigation Table

Scenario

Display irrigation recommendations for multiple plots.

Requirements

Create a table with columns:

- Plot ID
- Farmer Name
- Soil Moisture (%)
- Weather
- Irrigation Recommendation
- Water Requirement

Use

- <caption>
- <thead>
- <tbody>
- <tfoot>
- rowspan
- colspan

Apply suitable formatting.

Concepts Tested

- HTML Tables
- Table Attributes

Deliverable: irrigation-table.html

Task 7: Farmer Registration Form

Scenario

Create a registration page for farmers.

Requirements

Use form controls:

- Name
- Mobile Number
- Email
- Plot ID
- Village
- Crop Stage
- Soil Type
- Irrigation Method
- Date
- File Upload (Farm Photo)

Include

- <label>
- <fieldset>
- <legend>

Use HTML5 attributes

- required
- placeholder
- pattern
- autofocus

Concepts Tested

- HTML Forms
- HTML5 Input Types
- Validation Attributes

Deliverable: registration.html

Task 8: Display Weather Forecast using IFrame

Scenario

Display weather information without leaving the website.

Requirements

Embed

- IMD Weather Forecast page
- Google Maps showing the farm location

Use

- <iframe>

Place inside a section called

Today's Weather Forecast

Concepts Tested

- Inline Frames
- External Content Integration

Deliverable: weather.html

Task 9: Add Multimedia & Executable Content

Scenario

Create a multimedia awareness page for farmers.

Requirements

Add

- A video explaining Smart Irrigation
- Background audio explaining water conservation
- JavaScript alert

Welcome to the Irrigation Advisory System

Use

- <video>
- <audio>
- <script>

Enable

- controls
- autoplay
- loop

Concepts Tested

- Multimedia Tags
- JavaScript
- Executable Content

Deliverable: media.html

Code :

Task 1-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Smart Irrigation Advisory System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <section>
    <h2>About</h2>
    <p>This portal provides irrigation advisory services for sugarcane farms using AI and sensor data.</p>
  </section>

  <article>
    <h2>How It Works</h2>
    <p>Sensors collect soil and weather data, which is analyzed by an AI model to recommend irrigation
schedules.</p>
  </article>

  <aside>
    <h2>Note</h2>
    <p>This system is designed for farmers, agricultural officers, researchers, and sugar factory
management.</p>
  </aside>

  <footer>
    <p>&amp;copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
</html>
```

Task 2-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Farm Information</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <h2>Irrigation Workflow</h2>
  <ol>
    <li>Collect soil moisture and weather data</li>
    <li>Send data to <b>AI Advisory Engine</b></li>
    <li>Analyze data and predict irrigation requirement</li>
    <li>Generate irrigation recommendation</li>
    <li>Notify the <i>farmer</i></li>
  </ol>

  <h2>Sensors Used</h2>
  <ul>
    <li>Soil Moisture Sensor</li>
    <li>Temperature Sensor</li>
    <li>Humidity Sensor</li>
    <li>Rainfall Sensor</li>
  </ul>

  <h2>Crop Stages</h2>
  <dl>
    <dt>Germination</dt>
    <dd>Seed sprouting and initial root development.</dd>

    <dt>Tillering</dt>
    <dd>Formation of multiple shoots from the base of the plant.</dd>

    <dt>Grand Growth</dt>
    <dd>Rapid stem growth stage with <mark>highest water requirement</mark>.</dd>

    <dt>Maturity</dt>
    <dd>Sugar accumulation stage with reduced irrigation needs.</dd>
```

```
  </dl>

  <footer>
    <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
</html>
```

Task 3-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Irrigation Advisory</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <h2>Today's Irrigation Recommendation</h2>

  <p><strong>Plot ID:</strong> KJS-AGR-01</p>
  <p><strong>Farmer Name:</strong> Ramesh Patil</p>
  <p><strong>Soil Moisture:</strong> <mark>32%</mark></p>
  <p><strong>Recommended Irrigation Duration:</strong> <em>2 hours 30 minutes</em></p>
  <p><strong>Water Stress Level:</strong> <mark>Moderate</mark></p>

  <hr>

  <p>
    It is <strong>recommended</strong> to irrigate this plot within the next
    <em>24 hours</em>.<br>
    <small>Advisory generated on 02-08-2026</small>
  </p>

  <footer>
    <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
</html>
```

Task 4-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Farm Gallery</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <h2>Farm Gallery</h2>

  <a href="#healthy-crop">
    <figure>
      <img src="sugarcane.jpg" alt="Healthy Sugarcane Crop" title="Healthy Sugarcane Crop" width="300"
height="200">
      <figcaption>Healthy Sugarcane Crop</figcaption>
    </figure>
  </a>

  <a href="#drip-irrigation">
    <figure>
      <img src="irrigation-system.jpg" alt="Drip Irrigation System" title="Drip Irrigation System"
width="300" height="200">
      <figcaption>Drip Irrigation System</figcaption>
    </figure>
  </a>

  <a href="#soil-sensor">
    <figure>
      <img src="soil-moisture.jpg" alt="Soil Moisture Sensor" title="Soil Moisture Sensor" width="300"
height="200">
      <figcaption>Soil Moisture Sensor</figcaption>
    </figure>
  </a>

  <a href="#weather-station">
    <figure>
      <img src="weather-monitor.jpg" alt="Weather Monitoring Station" title="Weather Monitoring Station"
width="300" height="200">
```

```
    <figcaption>Weather Monitoring Station</figcaption>
   </figure>
  </a>

  <h2>Details</h2>
  <p id="healthy-crop"><b>Healthy Sugarcane Crop:</b> Sugarcane fields are large-scale agricultural plantations dedicated to growing tall, perennial grass species primarily for sugar and ethanol production. These fields thrive in tropical and subtropical regions, requiring abundant sunlight, warm temperatures, and heavy rainfall or robust irrigation to mature. Because the crop grows densely and can reach heights of up to several metres, the fields form massive, lush green canopies across the landscape. They require deep, well-drained fertile soil and are typically managed using structured row layouts to accommodate mechanical harvesting or manual cutting.</p>
  <p id="drip-irrigation"><b>Drip Irrigation System:</b> A drip irrigation system is an efficient watering method that delivers water directly to plant roots through a network of tubes and emitters. By applying water slowly and precisely at the soil surface, it drastically reduces evaporation and runoff compared to traditional sprinklers. This targeted delivery optimizes plant growth by keeping the root zone consistently moist without waterlogging. It is highly versatile, making it ideal for home gardens, greenhouses, and commercial agriculture. </p>
  <p id="soil-sensor"><b>Soil Moisture Sensor:</b> A soil moisture sensor is an electronic device that measures the volume of water held within soil particles. It typically functions by tracking either electrical resistance or capacitance to determine how wet or dry the ground is. These sensors provide crucial data that helps automated irrigation systems turn water pumps on or off precisely when needed. By preventing both underwatering and overwatering, they actively conserve water resources while keeping plants healthy. They are widely used by hobbyists using microcontrollers like Arduino, as well as by commercial farmers practicing smart agriculture.</p>
  <p id="weather-station"><b>Weather Monitoring Station:</b> A weather monitoring station is a facility or system equipped with specialized sensors and data loggers used to measure, record, and transmit atmospheric and environmental conditions. These systems track vital metrics like temperature, humidity, wind speed, solar radiation, and precipitation.</p>

  <footer>
   <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
</html>
```

Task 5-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Farm Plot Map</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
   <h1>Smart Irrigation Advisory System</h1>
   <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
   <a href="index.html">Home</a>
```

```
  <a href="farm-info.html">Farm Dashboard</a>
  <a href="advisory.html">Irrigation Advisory</a>
  <a href="irrigation-table.html">Sensor Data</a>
  <a href="registration.html">Contact</a>
 </nav>

 <h2>Farm Plot Navigation</h2>

 <img src="farm-map.png" alt="Farm Layout Map" width="600" height="400" usemap="#farmmap">

 <map name="farmmap">
  <area shape="rect" coords="0,0,200,400" href="#plot-a" alt="Plot A">
  <area shape="rect" coords="200,0,400,400" href="#plot-b" alt="Plot B">
  <area shape="rect" coords="400,0,600,400" href="#plot-c" alt="Plot C">
 </map>

 <h2>Plot Details</h2>
 <p id="plot-a"><b>Plot A:</b> Area 2.5 acres, Crop Stage: Tillering</p>
 <p id="plot-b"><b>Plot B:</b> Area 3.0 acres, Crop Stage: Grand Growth</p>
 <p id="plot-c"><b>Plot C:</b> Area 1.8 acres, Crop Stage: Maturity</p>

 <footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
 </footer>

</body>
</html>
```

Task 6-

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Irrigation Table</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>

 <header>
  <h1>Smart Irrigation Advisory System</h1>
  <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
 </header>

 <nav>
  <a href="index.html">Home</a>
  <a href="farm-info.html">Farm Dashboard</a>
  <a href="advisory.html">Irrigation Advisory</a>
  <a href="irrigation-table.html">Sensor Data</a>
  <a href="registration.html">Contact</a>
 </nav>

 <table>
  <caption>Plot-wise Irrigation Advisory</caption>
  <thead>
```

```
    <tr>
      <th>Plot ID</th>
      <th>Farmer Name</th>
      <th>Soil Moisture (%)</th>
      <th>Weather</th>
      <th>Irrigation Recommendation</th>
      <th>Water Requirement</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>KJS-AGR-01</td>
      <td>Ramesh Patil</td>
      <td>32%</td>
      <td>Sunny</td>
      <td>Irrigate Today</td>
      <td>50 mm</td>
    </tr>
    <tr>
      <td>KJS-AGR-02</td>
      <td>Sunita More</td>
      <td>45%</td>
      <td rowspan="2">Partly Cloudy</td>
      <td>Irrigate Tomorrow</td>
      <td>35 mm</td>
    </tr>
    <tr>
      <td>KJS-AGR-03</td>
      <td>Vikas Jadhav</td>
      <td>58%</td>
      <td>No Irrigation Needed</td>
      <td>0 mm</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="5">Total Water Requirement</td>
      <td>85 mm</td>
    </tr>
  </tfoot>
 </table>

 <footer>
   <p>&copy; 2026 Smart Irrigation Advisory System</p>
 </footer>

</body>
</html>
```

Task 7-

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
```

```
  <title>Farmer Registration</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <h2>Farmer Registration Form</h2>

  <form action="#" method="post">
    <fieldset>
      <legend>Farmer Details</legend>

      <label for="name">Name:</label><br>
      <input type="text" id="name" name="name" placeholder="Enter your name" required
autofocus><br><br>

      <label for="mobile">Mobile Number:</label><br>
      <input type="tel" id="mobile" name="mobile" placeholder="10-digit mobile number" pattern="[0-
9]{10}" required><br><br>

      <label for="email">Email:</label><br>
      <input type="email" id="email" name="email" placeholder="example@mail.com" required><br><br>

      <label for="plotid">Plot ID:</label><br>
      <input type="text" id="plotid" name="plotid" placeholder="e.g. KJS-AGR-01" required><br><br>

      <label for="village">Village:</label><br>
      <input type="text" id="village" name="village" placeholder="Enter village name" required><br><br>

      <label for="cropstage">Crop Stage:</label><br>
      <input type="text" id="cropstage" name="cropstage" placeholder="e.g. Tillering" required><br><br>

      <label for="soiltype">Soil Type:</label><br>
      <input type="text" id="soiltype" name="soiltype" placeholder="e.g. Black Soil" required><br><br>

      <label for="irrigation-method">Irrigation Method:</label><br>
      <input type="text" id="irrigation-method" name="irrigation-method" placeholder="e.g. Drip Irrigation"
required><br><br>

      <label for="date">Date:</label><br>
      <input type="date" id="date" name="date" required><br><br>
```

```
    <label for="photo">Farm Photo:</label><br>
    <input type="file" id="photo" name="photo"><br><br>

    <input type="submit" value="Register">
   </fieldset>
  </form>

  <footer>
   <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
</html>
```

Task 8-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Weather Forecast</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
   <h1>Smart Irrigation Advisory System</h1>
   <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
   <a href="index.html">Home</a>
   <a href="farm-info.html">Farm Dashboard</a>
   <a href="advisory.html">Irrigation Advisory</a>
   <a href="irrigation-table.html">Sensor Data</a>
   <a href="registration.html">Contact</a>
  </nav>

  <section>
   <h2>Today's Weather Forecast</h2>

   <p>IMD Weather Forecast:</p>
   <iframe src="https://mausam.imd.gov.in/" width="500" height="300" title="IMD Weather
Forecast"></iframe>

   <p>Farm Location:</p>
   <iframe src="https://maps.google.com/maps?q=sugarcane%20farm&output=embed" width="500"
height="300" title="Farm Location"></iframe>
  </section>

  <footer>
   <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

</body>
```

```
</html>

Task 9-
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Multimedia Awareness</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Smart Irrigation Advisory System</h1>
    <p>Smart Irrigation using AI &amp;amp; Sensor Technology</p>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="farm-info.html">Farm Dashboard</a>
    <a href="advisory.html">Irrigation Advisory</a>
    <a href="irrigation-table.html">Sensor Data</a>
    <a href="registration.html">Contact</a>
  </nav>

  <h2>Smart Irrigation Awareness Video</h2>
  <video src="videos/smart-irrigation.mp4" width="400" controls>
    Your browser does not support the video tag.
  </video>

  <h2>Water Conservation Awareness Audio</h2>
  <audio src="audio/water-conservation.mp3" controls autoplay loop>
    Your browser does not support the audio tag.
  </audio>

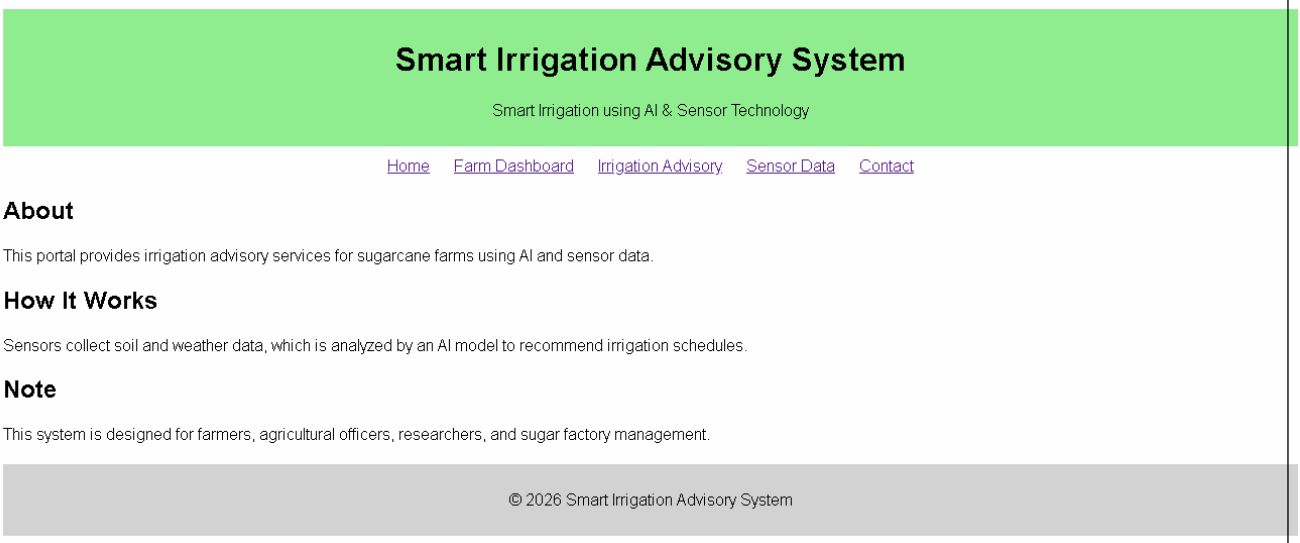
  <footer>
    <p>&copy; 2026 Smart Irrigation Advisory System</p>
  </footer>

  <script>
    alert("Welcome to the Irrigation Advisory System");
  </script>

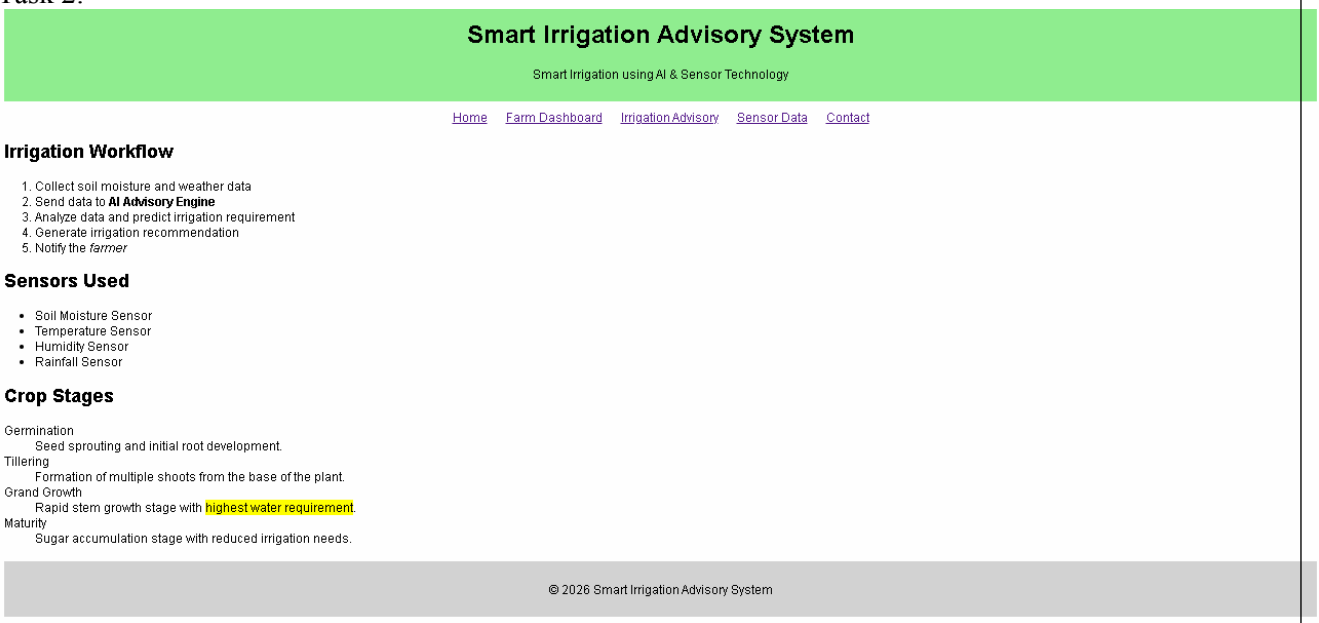
</body>
</html>
```

Output / Screenshot:

Task 1:-



Task 2:-



Task 3:-

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Home Farm Dashboard Irrigation Advisory Sensor Data Contact

Today's Irrigation Recommendation

Plot ID: KJS-AGR-01

Farmer Name: Ramesh Patil

Soil Moisture: 32%

Recommended Irrigation Duration: *2 hours 30 minutes*

Water Stress Level: Moderate

It is **recommended** to irrigate this plot within the next *24 hours*.

Advisory generated on 02-08-2026

© 2026 Smart Irrigation Advisory System

Task 4:-

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Home Farm Dashboard Irrigation Advisory Sensor Data Contact

Farm Gallery



Healthy Sugarcane Crop



Drip Irrigation System



Soil Moisture Sensor



Weather Monitoring Station

Details

Healthy Sugarcane Crop: Sugarcane fields are large-scale agricultural plantations dedicated to growing tall, perennial grass species primarily for sugar and ethanol production. These fields thrive in tropical and subtropical regions, requiring abundant sunlight, warm temperatures, and heavy rainfall or robust irrigation to mature. Because the crop grows densely and can reach heights of up to several metres, the fields form massive, lush green canopies across the landscape. They require deep, well-drained fertile soil and are typically managed using structured row layouts to accommodate mechanical harvesting or manual cutting.

Drip Irrigation System: A drip irrigation system is an efficient watering method that delivers water directly to plant roots through a network of tubes and emitters. By applying water slowly and precisely at the soil surface, it drastically reduces evaporation and runoff compared to traditional sprinklers. This targeted delivery optimizes plant growth by keeping the root zone consistently moist without waterlogging. It is highly versatile, making it ideal for home gardens, greenhouses, and commercial agriculture.

Soil Moisture Sensor: A soil moisture sensor is an electronic device that measures the volume of water held within soil particles. It typically functions by tracking either electrical resistance or capacitance to determine how wet or dry the ground is. These sensors provide crucial data that helps automated irrigation systems turn water pumps on or off precisely when needed. By preventing both underwatering and overwatering, they actively conserve water resources while keeping plants healthy. They are widely used by hobbyists using microcontrollers like Arduino, as well as by commercial farmers practicing smart agriculture.

Weather Monitoring Station: A weather monitoring station is a facility or system equipped with specialized sensors and data loggers used to measure, record, and transmit atmospheric and environmental conditions. These systems track vital metrics like temperature, humidity, wind speed, solar radiation, and precipitation.

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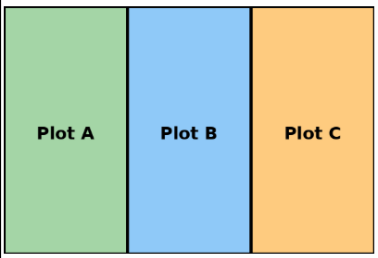
Task 5:-

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Home Farm Dashboard Irrigation Advisory Sensor Data Contact

Farm Plot Navigation



Plot Details

Plot A: Area 2.5 acres, Crop Stage: Tillering

Plot B: Area 3.0 acres, Crop Stage: Grand Growth

Plot C: Area 1.8 acres, Crop Stage: Maturity

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Task 6:-

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

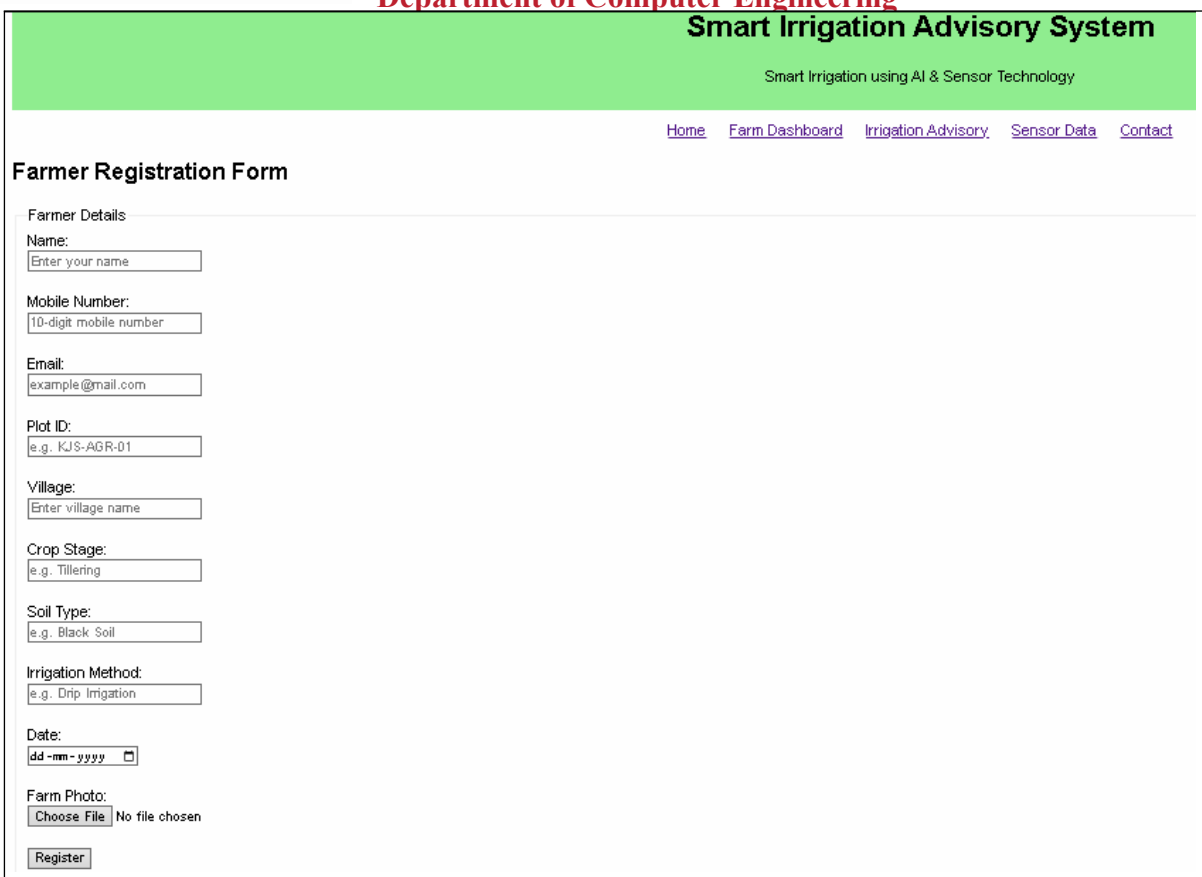
Home Farm Dashboard Irrigation Advisory Sensor Data Contact

Plot-wise Irrigation Advisory

Plot ID	Farmer Name	Soil Moisture (%)	Weather	Irrigation Recommendation	Water Requirement
KJS-AGR-01	Ramesh Patil	32%	Sunny	Irrigate Today	50 mm
KJS-AGR-02	Sunita More	45%	Partly Cloudy	Irrigate Tomorrow	35 mm
KJS-AGR-03	Vikas Jadhav	58%		No Irrigation Needed	0 mm
Total Water Requirement					85 mm

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Task 7:-


Smart Irrigation Advisory System
Smart Irrigation using AI & Sensor Technology
Home
Farm Dashboard
Irrigation Advisory
Sensor Data
Contact
Farmer Registration Form
Farmer Details
Name:
Enter your name
Mobile Number:
10-digit mobile number
Email:
example@mail.com
Plot ID:
e.g. KJS-AGR-01
Village:
Enter village name
Crop Stage:
e.g. Tillering
Soil Type:
e.g. Black Soil
Irrigation Method:
e.g. Drip Irrigation
Date:
dd-mm-yyyy
Farm Photo:
Choose File
No file chosen
Register

Task 8:-

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact

Today's Weather Forecast

IMD Weather Forecast:

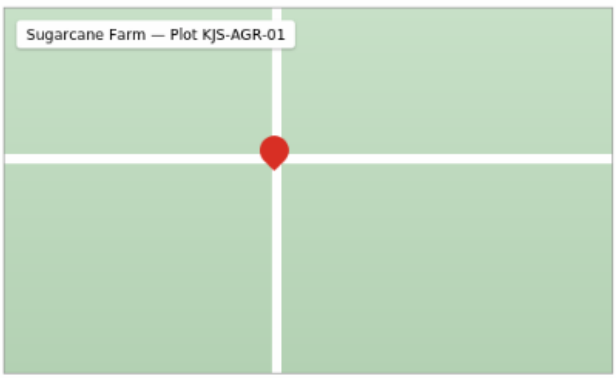
India Meteorological Department

Pune, Maharashtra

29°C

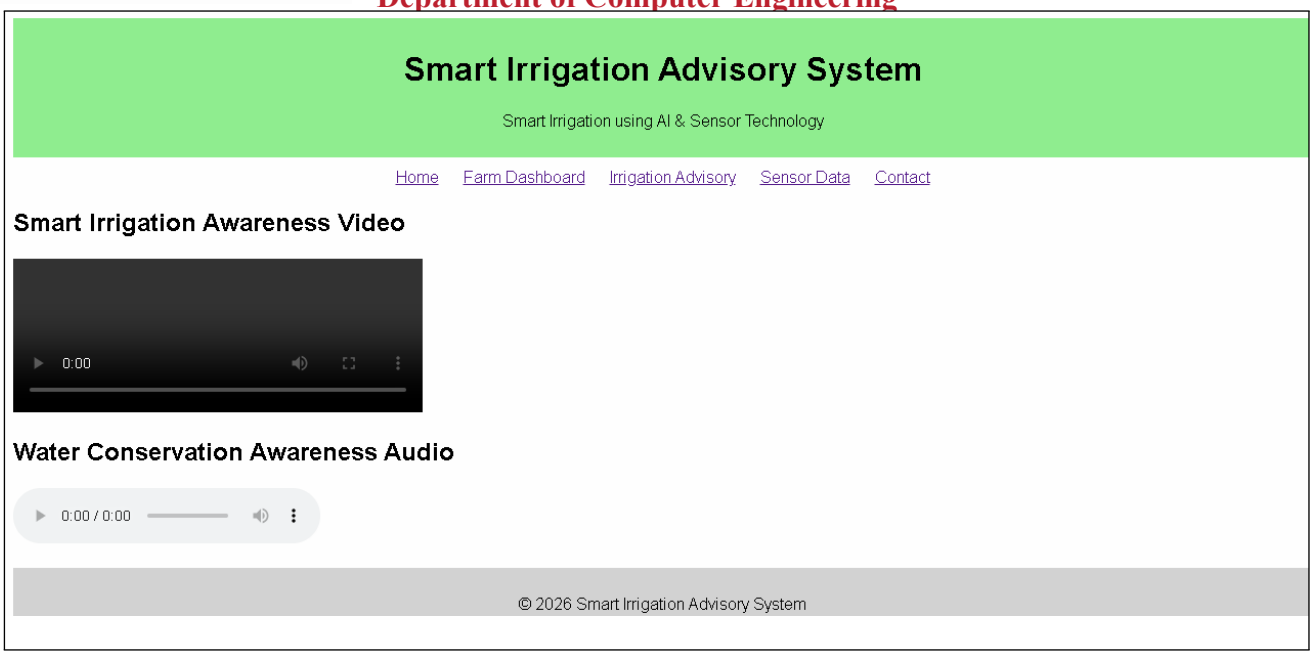
Condition:	Partly Cloudy
Humidity:	68%
Rainfall (24h):	4 mm
Wind:	12 km/h SW

Farm Location:



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Task 9:-



Conclusion and Discussion:

Across all tasks, core HTML5 principles were implemented to build interactive, well-structured, and accessible clinical web pages. Key concepts covered include interactive image maps, styled semantic data tables with cell merging, validated patient registration forms, embedded external iframes, and controlled multimedia integration with JavaScript. All deliverables (lab-map.html, test-price-table.html, patient-registration.html, health-information.html, and media.html) fully satisfy the academic requirements for the Web Development Laboratory.